Surveillance & Society

Essay

HOMECTRL: An Example of Practice-Based Artistic Research as a Methodology in Surveillance Studies

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Abstract

This essay provides an example of how practice-based artistic research can contribute to surveillance studies. It does so by reflecting on the development of the installation HOMECTRL for The Danish Museum of Science & Technology. The installation offers speculative fabulations regarding how the Internet of Things (IoT) is rearticulating and transforming the home as a site of control, on a spectrum ranging from the mundane everyday use of robot vacuum cleaners to domestic abuse facilitated by IoT technologies. By sharing excerpts from narrative monologues that probe the question of where control is located, and still images that experiment with representing how technologies perceive their surroundings, the essay demonstrates that practice-based methods can open up understandings of the home as a distributed spatial structure permeated by smart technologies that have new abilities to access, control, and manipulate domestic space.

Introduction

Over the last twenty years, ambient digitised forms of surveillance have occupied an increasingly predominant position in surveillance practices. Indeed, we now might even speak of an overload (McCullough 2013) or calmness (Weiser 1996) of surveillance, whereby surveillance paradoxically blends into the background of everyday life precisely because it is everywhere (Wellendorf, Søilen, and Veel 2022). This constant monitoring attracts our attention mainly when it produces obstacles to the continuation of the everyday and its rhythms. In parallel with this increasing ubiquity of surveillance, art and cultural studies have gained significance in the study of surveillance techniques and practices. Although surveillance has been a coveted topic of authors, visual artists, and film directors throughout the twentieth century, the everyday embeddedness of surveillance technologies today means that it forms part of the backdrop for stories and artworks whose primary topic is not surveillance per se. This broadens the scope of "surveillance

[1] The HomeCTRL installation is developed by media artist Kassandra Wellendorf and cultural studies scholar Kristin Veel. For the conception of the visual vocabulary, we solicited the expertise of Copenhagen-based Director of Photography, Mia Mai Dengsø Graabæk.

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art" and draws attention to how art and cultural disciplines are particularly attuned to the ephemeral ambiance—the atmosphere (Böhme 2016) or *Stimmung* (Gumbrecht 2012)—of surveillance (Søilen 2020). Important work has been done in terms of showing how analyses of film, art, and literature provide insights into surveillance discourses (Kammerer 2008; Marks 2015; Søilen 2021; Wiehn 2022), and the significance of the critical perspectives that art brings to surveillance is increasingly recognised (Stark and Crawford 2019; Thylstrup et al. 2021), not least with the inauguration of the Surveillance Studies Network Arts Prize.

In this essay, we explore how art can contribute to the study of surveillance, not just as an object of scholarly analysis or a distinctive form of knowledge production but as a practice-based research methodology. We describe the process of using speculative fabulations to explore the spectrum from everyday to abusive uses of smart technologies in the home during the development of our interactive installation *HOMECTRL* for the The Danish Museum of Science & Technology. With regard to feminist speculative fabulation, Donna Haraway (2013) notes: "It matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with; it matters what knots knot knots, what thoughts think thoughts, what ties tie ties. It matters what stories make worlds, what worlds make stories." By reflecting on our methodology, this essay foregrounds "what matters we use to think other matters with" in the development of our art installation, its narrative voice, and its visual vocabulary.

CTRL + Home

The experience of the technology-saturated home is central to *HOMECTRL*. By 2030 there will be an estimated 125 billion Internet of Things (IoT) devices worldwide (IHS Markit 2017). Many of these devices are found in domestic settings. Door locks, interactive assistants, surveillance systems, and sensors that chart humidity, heat, and light are Wi-Fi enabled and can be controlled via smartphone apps. The home thus becomes portable and its boundaries porous, in new ways we have conceptualised elsewhere as "leaky" (Wellendorf, Søilen, and Veel 2022). The home is not a bounded, contained, static space; it is experienced as decentred, spatially distributed, and lacking clear boundaries between private and public, work and leisure, safe and unsafe (Harris 2018; Woodlock 2017). Notably, thanks to smartphones, the home and its control are distributed across multiple sites and platforms.

HOMECTRL works with the concepts of spatial distribution, porosity, and leakiness. Its "homely" set design includes a comfy chair with a smartphone and earphones on it. Museum visitors are invited to pick up the phone and put on the earphones through which narrative voices guide them into a number of homes. The narrative voices address listeners directly, encouraging them to identify with different characters who display various forms of attachment and addiction to smart technology.



Figure 1: *Photo by Kristin Veel.*

You sink into the chair. The mobile phone weighs heavily in your hand. You close your eyes and dive into the darkness behind your eyes. A moment of rest. You start to attend to your own breathing. But then the phone draws your attention again. You look at the icons on the screen. Look at them. Touch me! Touch me! No, me! It's so tempting to touch something that lights up. It's as if it's not you who is making the decision. "Touch me," whispers the little green radar symbol with the blue dot in the middle. It's there in order for you to touch it. You touch it. Squeeze it under your finger as if to keep it from disappearing.

Feminist theory has emphasised the ambivalence of the home as simultaneously a sanctuary and a site of care duties, labour, abuse, and violence (Cockburn and First-Dilić 1994). This ambivalence is key to *HOMECTRL*, a central premise of which is that conceptions of the home are predicated on gendered, cultural, and historical factors, and that the integration of IoT technologies can be regarded as the most recent iteration of the home's ambiguity. Domestic technologies are frequently marketed in terms of progress, efficiency, convenience, and safety, and they can even provide protection in cases of abuse. But they also offer new tools for old controlling behaviours and abuse within the confines of the home (Tanczer et al. 2018). Victims report flickering lights, locking doors, and blaring music that wakes them at night, often controlled by ex-partners who retain access to the home through connected devices (Bowles 2018; Small 2019). However, *HOMECTRL* explores not only such extreme cases but also how the affordances of these technologies can trigger controlling behaviour in mundane situations. Thus, it prompts the audience to ask how far along the spectrum they can identify and empathise.



Figure 2: *Photos by: Mia Mai Dengsø Graabæk and Kristin Veel.*

To chart the interactions between bodies and technologies, we developed an experimental methodology that draws on participatory visual methodology (Mitchell, De Lange, and Moletsane 2017), ethnomethodology (Garfinkel 1989; Wellendorf 2015), sensory ethnography (Pink 2015), experience sampling (Larson and Csikszentmihalyi 2014), and design anthropology (Gunn, Otto, and Smith 2013). We conducted a number of interviews with informants who had experience of living with, controlling, or being controlled by IoT technologies in the home—ranging from everyday observations of robot vacuum cleaners to severe cases of tech abuse.

We filmed the interviews not only to gather information but also as a generative process that informed the development of our conceptualisations, visual vocabulary, and narrative voices. The visual vocabulary we developed for the installation speculatively probes the questions: How do IoT technologies perceive us and our homes? Where is control located? To answer these questions, we recreate the surveillance technologies' own perspectives in various experimental ways (e.g., by holding filters in front of a camera or by creating low-tech arrangements such as a tea towel with a mobile phone on top to emulate the movement of a robot vacuum cleaner). While the visual vocabulary seeks to translate affective experiences, the auditory component—the narrative voices—solicits audience identification and presents ethical conundrums. Each of the narratives features a fictional character presented as a "you." This presentation invites the audience to identify with the characters' varying degrees of emotional attachment and addiction to connected devices, and to sense the slippage from care to control that the technologies incite. In this way, the short narratives explore how the technologies' affordances can trigger psychological control mechanisms that lead to actions that bypass our ethical standards. The installation stages four different scenarios connected to different technologies: (1) a robot vacuum cleaner; (2) a baby monitor and a smart sock that tracks heart rate and oxygen levels; (3) a mobile pet camera; and (4) domestic abuse enabled by IoT technologies. The following gives examples of the four scenarios' visual vocabularies and narrative voices.

From Dust You Came

Scenario: You have recently purchased a robot vacuum cleaner, which you can control from your phone. It has a camera too, so you can keep an eye on your home. While you are at work, you keep checking its movements—looking out for dust. However, it is almost impossible to spot dust and dirt through the camera, and since the camera points towards the ceiling, you cannot watch out for dust on the floor. But you imagine you can see it anyway. It is as if you can see the dust piling up around the vacuum cleaner as it moves about. Gradually, your obsession with cleaning grows, and you start leaving crumbs in corners to test whether the robot will find them.

Vocabulary: Using a robot vacuum cleaner (Trifo Max) with a built-in surveillance camera, we created imagery that visualises the home from the robot's perspective, moving along the floorboards and looking up at the ceiling. The images change constantly from high resolution to grainy and pixelated. Consequently, the solidity of the home's four protective walls dissolves, visualising the porosity and leakiness of the smart home as it appears to liquefy and spill out of the frame. While the high-resolution imagery shows you potential dirt in odd places, the pixelated images make your home difficult to recognise; the dust becomes part of the image, dirtying your vision of your home.

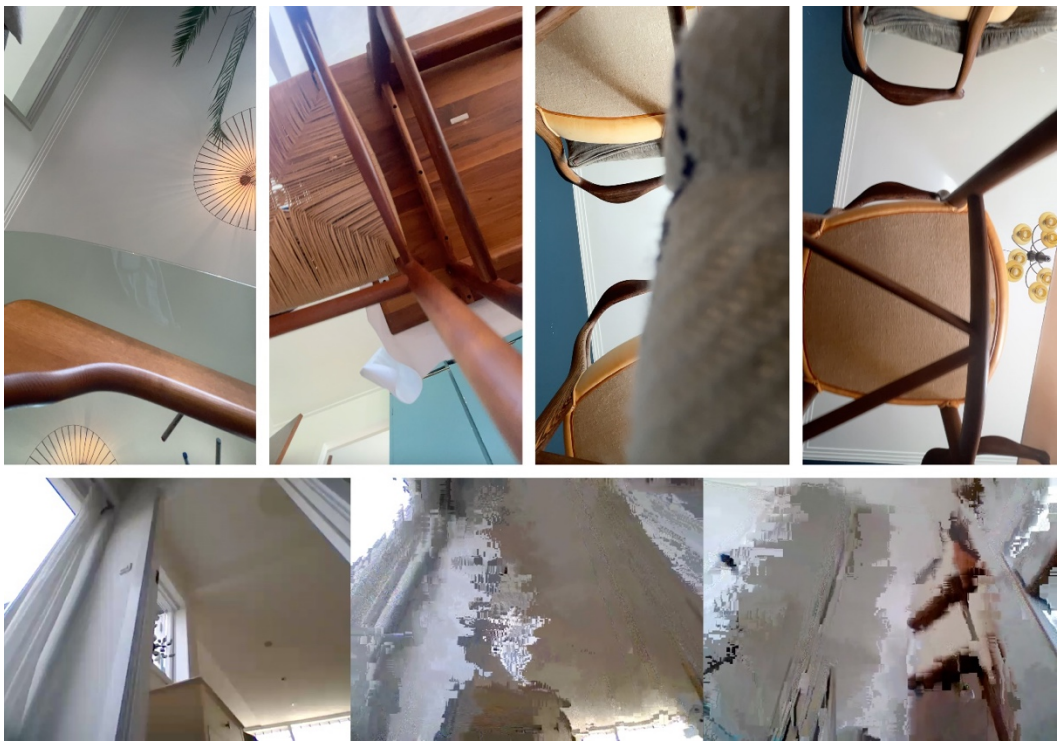


Figure 3: *Photos by Mia Mai Dengsø Graabæk.*

Your thoughts start drifting back to the robot at home. Is Robertus still cleaning the kitchen? Did he find the crumbs you placed in the tricky spot between the dining table and the fridge? A little test to show how much you love him. You log on to the app. It feels nice to see your living room through his lens, almost as if you're home already. Even though the angle is so different from the way you normally perceive your home. It feels familiar and strange at the same time. You see parts of your home you've never thought about before. You notice cobwebs and insects in ceiling corners, greasy fingermarks or food stains on walls. Is that a piece of chewing gum under that chair? You start making notes on a piece of paper, even drawing a map of your home as seen from below, so you can remember where to find the stains and remove them when you get home. It feels as if you are a team, you and the robot.

Mama's Gonna Buy You a Looking Glass

Scenario: You monitor your child through a camera as well as a smart sock that measures heart rate and oxygen levels. This caring activity fuels your anxiety that the child might die in its sleep.

Vocabulary: To visualise the surveillance of a baby in its cot, we use various coloured and wrinkled filters to evoke a mixture of baby monitor images, foetus ultrasound imagery, and cultural references from films such as the 1982 film *Poltergeist*, all of which contribute to a cultural imaginary concerning the intermingling of technology and child protection. The intention is to visually evoke the anxiety and ambivalence involved in using technology to care for the vulnerable.

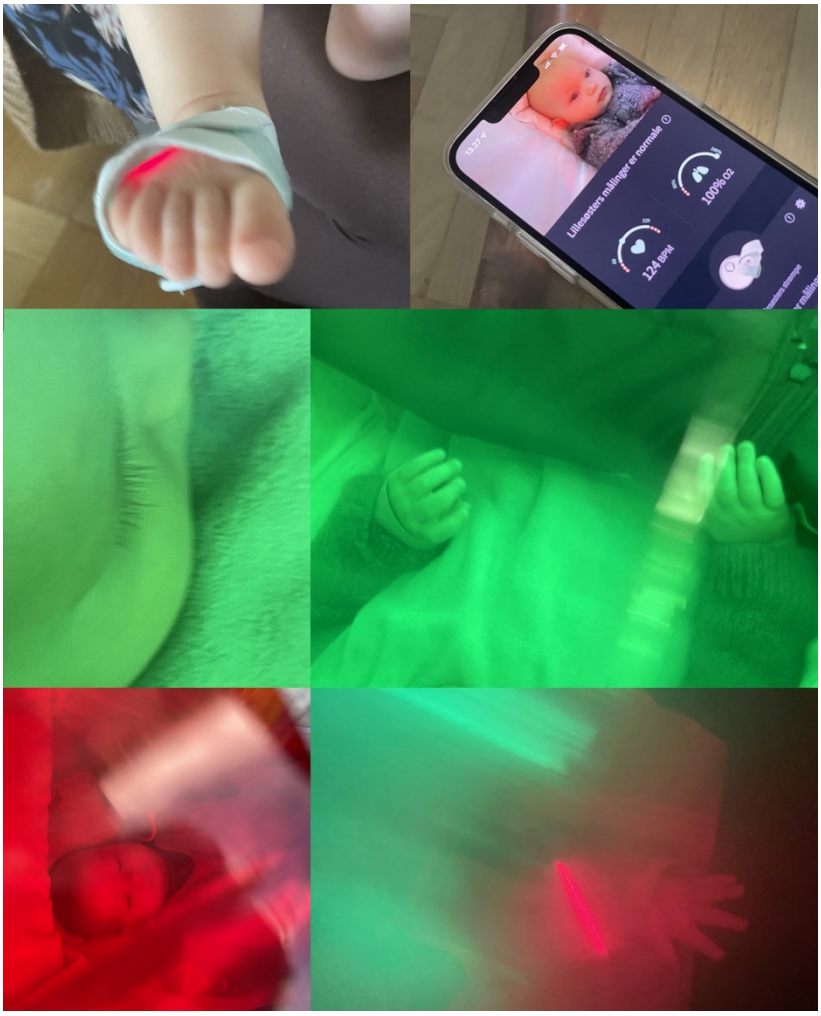


Figure 4: *Photos by Mia Mai Dengsø Graabæk.*

You stare at the image on the screen. Everything is ok. It's just a healthy sleeping child. Of course she's breathing. Also, you have the smart sock, which measures her heart rate and oxygen level. It would let you know if something was wrong. It would be good if you could sleep too. You close your eyes and feel sleep dragging you down.… Bip-bip-bip… What was that? Did you dream it, or was there an alarm? The app tells you her heart rate and oxygen level are normal. You really need to relax and stop checking the app all the time. You look closely at the image. Isn't she lying in an awkward position? Almost too peaceful? Is her chest moving ever so slightly up and down? If you squint hard, it seems as if she can somehow tell through the screen that she should move a limb or shift her position to calm you. There. A tiny finger moved. Phew.

Hide and Seek

Scenario: You have a mobile robot camera that allows you to see if your cats are alright when you are not at home. You take great pleasure in watching how the animals behave when you are not around. When you go away for a weekend, and you ask a neighbour to look after the cats, you can also watch what the neighbour does inside your home.

Vocabulary: The small robot camera is agile and can be precisely controlled. It can drive under things, jump, turn around, and make somersaults. A perfect little spy. We filmed through the robot's own camera, showing it playing with the cats and simultaneously spying on people from hidden corners. The hide-and-seek camera movements translate the complex combination of the desire to spy on others and the increasing feeling of shame. We thus visualise the slippery slope from socially accepted uses of technology to ethical dilemmas fed by the addictions the technology affords. For this purpose, we lit the images partly with the robot's own red laser, which can be used to play with cats; we also lit the robot with a nanny cam's night vision feature.

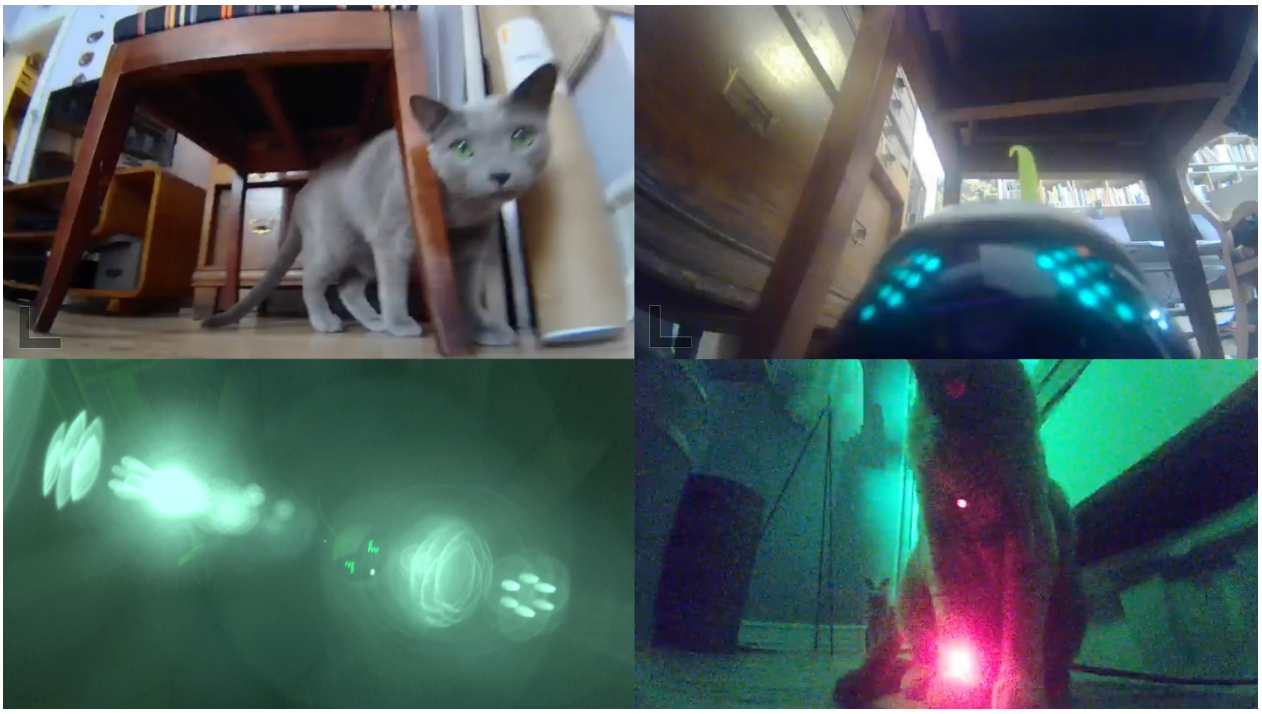


Figure 5: *Photos by Mia Mai Dengsø Graabæk.*

It feels as if you extend your body into the robot. As if your eyes are roller-skating beneath all your furniture. Playing hide and seek with your cats. It has a loudspeaker so you can talk to them, but they don't really react. You use it more and more often. When you go out for a walk, when you go down to the beach, when you go shopping. It's nice to know what's happening inside your home. It's like following a long, slow TV series. Sometimes you see your neighbours who look after your cats when you are away, and then you know you shouldn't be looking. But you leave the camera on anyway, watching them move within the space that is your home. When they move close to the robot, you duck as if they might see you, and your face reddens with shame. But you don't turn the camera off. It's your home, after all, and you want to see how they treat your cats. But it's weird and confusing to feel that you're trespassing in your own home.

Gaslighting

Scenario: Your relationship has ended, and you are now shut out from what used to be your shared home. However, you still have access to the IoT technologies that permeate it, and you start to harass your ex by activating the alarm, changing the room temperature, turning on the dishwasher late at night, and making the light bulbs flicker in order to take back control.

Vocabulary: The visual vocabulary is inspired by the term "gaslighting," which psychological literature defines as "presenting false information to the victim with the intent of making them doubt their own memory and perception" (Breiding et al. 2015: 15). Lopez-Neira et al. (2019) have linked gaslighting to the use of IoT in domestic abuse. However, the term originates from Patrick Hamilton's 1938 play *Gas Light*, a story about a wife whose husband manipulates her perception of reality. The installation images play with the different angles from which the home can be observed remotely through connected devices. A key challenge was to translate the blurring of the senses that can occur when a victim of domestic tech abuse starts to doubt their own perception. We used lighting from different surveillance devices (nanny cams, scouting cameras, etc.), experimenting with night vision modes and infrared light to steep the walls in paranoia. We also used the colour green overlaid with occasional red spots, and we created visual feedback effects by letting different surveillance technologies film and light one another.



Figure 6: *Photos by Mia Mai Dengsø Graabæk.*



Figure 7: *Photos by Mia Mai Dengsø Graabæk.*

You can see she has trouble sleeping. She's probably struggling to get used to your no longer being around. Getting used to turning over in bed in that obnoxious, noisy way that wakes you up every time. Time and again you told her to try to be quieter. Now you *can make sure there's noise, and she can find out what it's like not to be able to sleep. It's been weeks, and she hasn't been reacting much, but now you can see through the alarm system's surveillance camera that she jumps when the radiator makes clanking sounds at night because you've turned it off. In the mornings you can see her adjusting the thermostat, probably wondering why it's so cold. Then you turn it up to maximum. Sometimes you turn off her alarm clock, but she doesn't try to fix it. It almost annoys you that she doesn't realise it's you doing these things. You can see her internalising it. Believing that she's losing it. Believing it's her own ineptitude with technology. Perhaps she even misses you. Thinks about calling you to come and fix it. Perhaps she tells her friends about it. But they're as technically incompetent as she is. They will tell her it's stress from the break-up that's making her see and hear things. They'll ask her to take care of herself, go to bed early, and drink more chamomile tea. Ha!*

Concluding Remarks on Control and Speculative Fabulation

To explore how the integration of IoT technologies is rearticulating and transforming the home as a site of control, we considered a whole spectrum from mundane to abusive uses. The visual vocabulary and narrative voices developed in our four scenarios demonstrate that practice-based artistic research can foreground the often-invisible affects and emotions of contemporary surveillance practices and techniques. By narratively placing the audience in the position of technology users, and through our visual vocabulary, we seek to create an embodied sense of the home as a distributed spatial structure permeated by smart technologies. In this way, we invite the audience to reflect on their own motivations for using such technologies to access, control, and manipulate their own domestic spaces.

Design anthropologist Sarah Pink (2023: 110) notes that if connected smart devices are integrated into the home in order to measure, predict, and ultimately optimise how the home works, then this may be at odds with the way "householders might want to stay in control so that they can make contingent decisions on about when technologies are used." This pinpoints the difficulty of determining where *control* is situated when we integrate the home with connected devices that learn from and even predict our behaviour. As we have seen, the question of locating control is the prime driver behind *HOMECTRL*'s speculative fabulations. For Pink (2023: 109), the home is a "a site for anticipation and human creativity and improvisation, where people have innumerable ways in which to organise, direct, prepare for and imagine what will happen in their immediate and far futures." We share her interest in the home as a site of speculation due to its inherent ambivalences. Yet our scenarios show that human creativity may also play out at the manipulative end of the spectrum, and that control is a slippery concept when distributed across human and non-human entities. For us, art as a practice-based research methodology to generate speculative fabulations is a means to address this slipperiness, and to respond to Haraway's (2013) caution. "It matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with"; and we might add, it matters how we control the control devices we use to control our homes.

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